Prateek Jaiswal

Contact Information

Young Hall (420)
Daniels School of Business
Purdue University
West Lafayette, IN 47906 USA

Phone: +1 (765) 409-5910
E-mail: jaiswalp@purdue.edu
Homepage: pvjaiswal.github.io

Education

Ph.D., Industrial Engineering, Purdue University, West Lafayette, IN, USA **August 2021**

B.Tech., Mechanical Engineering, Indian Institute of Technology, Patna, India **May 2012**

Academic Positions

Clinical Assistant Professor **August 2024 - present**
Department of Quantitative Methods, Daniels School of Business, **Purdue University, West Lafayette**

Postdoctoral Research Associate **September 2021 - August 2024**
Department of Statistics, **Texas A&M University, College Station**

Research Interests

Methodology: Sequential Decision-Making Methods, Data-Driven Optimization, Bayesian Statistics, Interplay of Machine Learning, Optimization, and Statistics
Applications: Healthcare Operations Management, Precision Medicine, Applications of AI and ML for Social Impact

Honors and Awards

Top Reviewer Award, NeurIPS **2025**

Community in Curriculum Award, Daniels School of Business, Purdue University **2025**

Finalist, INFORMS Data Mining Best Paper Competition (General Track) **2022**

Joe Newton Best Poster Award, Conference on Advances in Data Science: Theory, Methods and Computation **2022**

Travel grants: Sustainable Horizons Institute (SIAM CSE21 and BE program); PGSG (INFORMS 2020, NeurIPS 2020) **2020–2021**

Healthcare Decision-Making

Learning When to Trust in Observational Data: Deconfounded Transfer for Online Decision-Making
Jaiswal, P., Keyvanshokooh, E., and Cao, J. *Submitted to Management Science, under review.*

Toward Learning-Based Decision Support for Kidney Allocation: Within-Program Candidate Selection Under Ranking Constraints and Confounding
Jaiswal, P., and Gupta, D. *Working paper.*

Learning to Graduate from Mobile Health Interventions: An Interpretable, Certified Optimal-Stopping Algorithm
Jaiswal, P.*, and Zhalechian, M.*. *Working paper.*

Sequential Learning under Uncertainty

Posterior Tempering Explains Variance Inflation in Linear and Generalized Linear Thompson Sampling
Jaiswal, P., Pati, D., Bhattacharya, A., and Mallick, B.K. *Revision 1 submitted to Mathematics of Operations Research.*

Generalized Regret Analysis of Thompson Sampling using Fractional Posteriors
Jaiswal, P., Pati, D., Bhattacharya, A., and Mallick, B.K. *Revision 2 submitted to Journal of*

Machine Learning Research.

Frequentist Regret Analysis of GP-TS via Fractional Posteriors
Roy, S.*, **Jaiswal, P.***, Pati, D., Bhattacharya, A., and Mallick, B.K. *Submitted to Stochastic Systems, under review.*

Leveraging Offline Data from Similar Systems for Online Linear Quadratic Control
Bajaj, S., **Jaiswal, P.**, and Gupta, V. *IEEE Transactions on Automatic Control*, forthcoming (March 2027 issue). **(TAC)**

Personalized and Collaborative Online LQR via Thompson Sampling
Bajaj, S., **Jaiswal, P.**, and Gupta, V. *40th Conference on Neural Information Processing Systems, 2026* (accepted). **(NeurIPS)**

Risk-Aware Optimization

On the Statistical Consistency of Risk-Sensitive Bayesian Decision-Making
Jaiswal, P., Honnappa, H., and Rao, V.A. *37th Conference on Neural Information Processing Systems, 2023.* **(NeurIPS)**

Bayesian Joint Chance Constrained Optimization: Approximations and Statistical Consistency
Jaiswal, P., Honnappa, H., and Rao, V.A. *SIAM Journal on Optimization*, 33(3):1968–1995, 2023. Shorter version in *Proceedings of the 2nd Symposium on Advances in Approximate Bayesian Inference*, PMLR 118:1–12, 2020. **(SIOPT)**

Multistart Algorithm for Identifying all Optima of a Nonconvex Stochastic Oracle
Jaiswal, P., and Larson, J. *Optimization Letters* 18:1335–1360, 2024. **(OL)**

Asymptotic Consistency of α-Rényi-Approximate Posteriors
Jaiswal, P., Rao, V.A., and Honnappa, H. *Journal of Machine Learning Research* 21(156):1–42, 2020. **(JMLR)**

Asymptotic Consistency of Loss-Calibrated Variational Bayes
Jaiswal, P., Honnappa, H., and Rao, V.A. *Stat* 9(1):e258, 2020. **(Stat)**

Statistical Inference for Approximate Bayesian Optimal Design
Jaiswal, P., and Honnappa, H. *Proceedings of the 2020 Winter Simulation Conference.* **(WSC)**

Estimating Stochastic Poisson Intensities Using Deep Latent Models
Wang, R., **Jaiswal, P.**, and Honnappa, H. *Proceedings of the 2020 Winter Simulation Conference.* **(WSC)**

Optimal Allocations for Sample Average Approximation
Jaiswal, P., Honnappa, H., and Pasupathy, R. *Proceedings of the 2018 Winter Simulation Conference.* **(WSC)**

* Equal contribution.

Grant Writing

Jaiswal, P. (PI), Jaiswal A. (Co-Pi), Asish A (Co-Pi); "AI-persona driven case studies for semiconductor workforce development", National Science Foundation, 2026 (Rejected but competitive)

Teaching Experience

Instructor, Daniels School of Business, Purdue University, West Lafayette, IN

MGMT306- Management Science	(Median Rating 4.5/5.0)
QM472- Advanced Spreadsheet Modeling and Simulation	(Median Rating 4.0/5.0)

MGMT590- AI-Enhanced Optimization (Median Rating 5.0/5.0)
DSB60110- Industry Practicum (Graduate)

Teaching Assistant, School of Industrial Engineering, Purdue University, West Lafayette
IE343-Engineering Economics Fall 2016
IE533-Industrial Applications of Statistics Spring 2017
IE336-Operations Research-Stochastic Models Fall 2020
IE230-Probability and Statistics in Engineering Spring 2021

Experience

Purdue University, West Lafayette, IN

Graduate Research Assistant, School of Industrial Engineering **May 2017 - May 2020**
Topics: Variational inference for stochastic programming, Large deviations analysis of Sample average approximation

Argonne National Laboratory, Lemont, IL

Givens Associate **May 2020 - Aug 2020**
Topic: Multistart stochastic optimization method to tune the parameters of a high-energy physics event generator.

Givens Associate **May 2019 - Aug 2019**
Topic: Multistart algorithm for non-convex stochastic optimization.

Bharat Petroleum Corporation Ltd. (BPCL), India

Assistant Manager, Engineering & Projects **Jul 2012 - Jul 2016**

University of Auckland, New Zealand

Research Assistant, Centre for Advanced Composites Manufacturing **May 2011 - Jul 2011**
Topic: Manufacturing and analysis of Graphene-based nano-composites.

Invited Talks

Jaiswal, P., "Bayesian methodologies for data-driven decision-making",
Department of Statistics, Indiana University, Bloomington, Dec 2023
Industrial & Systems Engineering Program, University of Cincinnati, Dec 2023
Department of Industrial and Systems Engineering, University of Tennessee, Knoxville, April 2024

Jaiswal, P., Honnappa, H., and Rao, V.A. "Variational Bayesian method for Stochastically Constrained System Design Problem", *The 13th Young European Queueing Theorists (YEQT) workshops, EURANDOM, TU Eindhoven, The Netherlands - Oct 2019.*

Jaiswal, P., Honnappa, H., and Rao, V.A. "Variational Bayes for Data-driven Newsvendor Problem", *Conference on Data Science for Business and Economics, Purdue University, West Lafayette, IN, USA - May 2018.*

Conferences & Workshops Presentations

Jaiswal, P., Keyvanshokooh, E., and Cao, J. "Learning When to Trust in Observational Data: Deconfounded Transfer for Online Decision-Making" - *INFORMS Annual Meeting, San Francisco, CA (upcoming) - Nov 2026.*
- INFORMS Healthcare Conference - July 2026.
- POMS Annual Conference - May 2026.

Jaiswal, P., Pati, D.; Bhattacharya, A.; and Mallick, B.K. "Generalized Regret Analysis of Thompson Sampling using Fractional Posteriors" - *Conference on Advances in Data Science: Theory, Methods and Computation - Oct 2022.*
- INFORMS Annual Meeting - Oct 2022.
- TTIC Summer Workshop: New Models in Online Decision Making for Real-World Applications - July 2022.

Jaiswal, P., Honnappa, H. "Statistical Inference for Approximate Bayesian Optimal Design", *Winter Simulation Conference (Virtual)- Dec 2020.*

Jaiswal, P., Honnappa, H. "Variational Inference for Bayes Optimal Design", *INFORMS Annual Meeting (Virtual) - Nov 2020.*

Jaiswal, P., Honnappa, H., and Rao, V.A. "Variational Inference for Risk-Sensitive Decision-Making", *NeurIPS Workshop on Safety and Robustness in Decision Making - Dec 2019.*

Jaiswal, P., Honnappa, H., and Rao, V.A. "Variational Bayesian method for Stochastically Constrained System Design Problem", *INFORMS Annual Meeting - Oct 2019.*

Academic Service

Academic Leadership and Committees

Daniels School of Business, Purdue University, West Lafayette, IN
Academic Director, B.S. Program — Fall 2025 – Spring 2026
Business Analytics and Information Management (BAIM)
Clinical Assistant/Associate Professors Search Committee Member — 2025

Academic Reviewer

Journals	*(manuscripts reviewed)*
Journal of Machine Learning Research (JMLR)	(2)
Journal of American Statistical Association (JASA)	(2)
IISE Transactions	(1)
Stochastic Systems	(2)
Optimization Letters	(3)

Conferences	
International Conference on Machine Learning (ICML)	(4)
Conference on Neural Information Processing Systems (NeurIPS)	(12)
International Conference on Artificial Intelligence and Statistics (AISTATS)	(4)
Uncertainty in Artificial Intelligence (UAI)	(1)
AAAI Conference on Artificial Intelligence (AAAI)	(5)

Member

Institute for Operations Research and the Management Sciences (INFORMS)
Production and Operations Management Society (POMS)